L'ORÉAL

FROM AI DISCOVERY TO MEASUREMENT

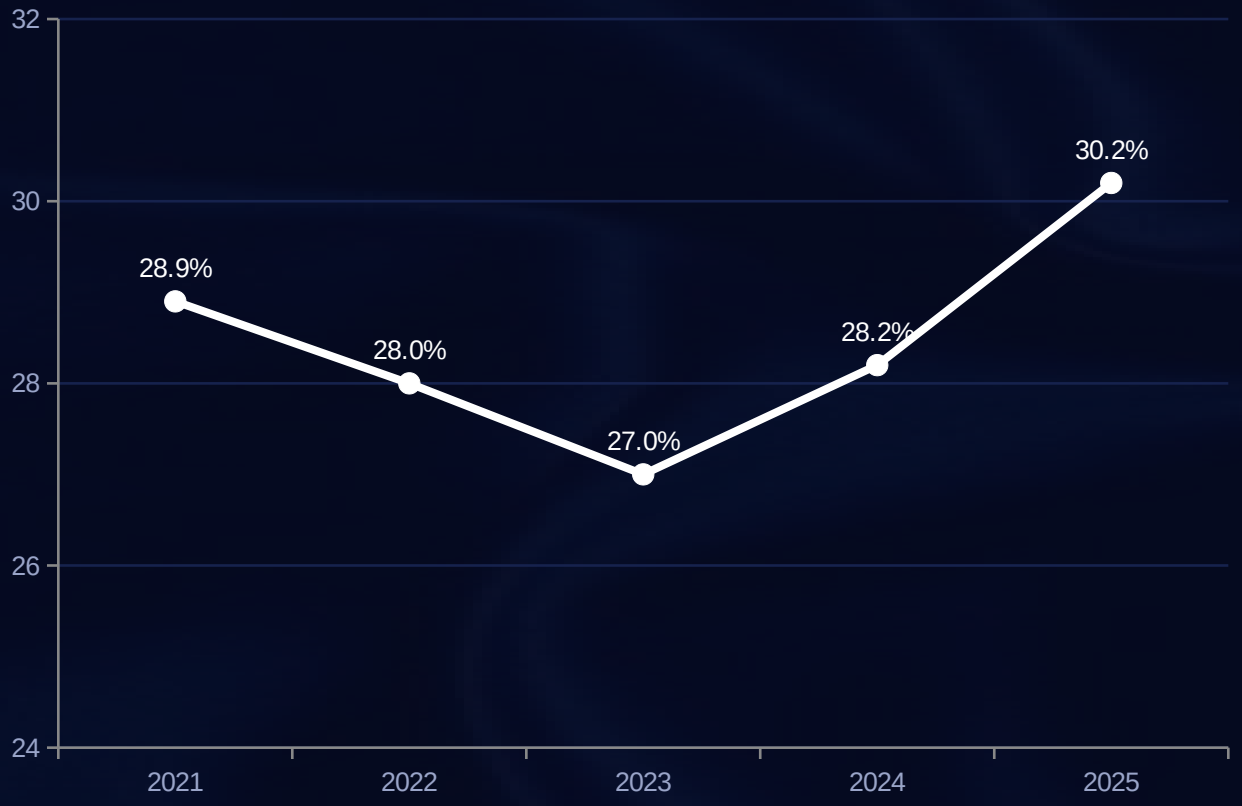
How L'Oréal could extend its existing measurement capabilities into conversational AI discovery — without needing to own the AI touchpoint.

L'ORÉAL · BUSINESS ANALYTICS & AI MEASUREMENT · 2026



E-COMMERCE SHARE OF SALES DIDN'T ALWAYS RISE

E-commerce share fell **1.9 pp from 2021–2023**, then recovered to 28.2% in 2024 and reached 30.2% in 2025.



The decline did not mean e-commerce sales were falling — total L'Oréal sales were growing faster in 2022 and 2023.

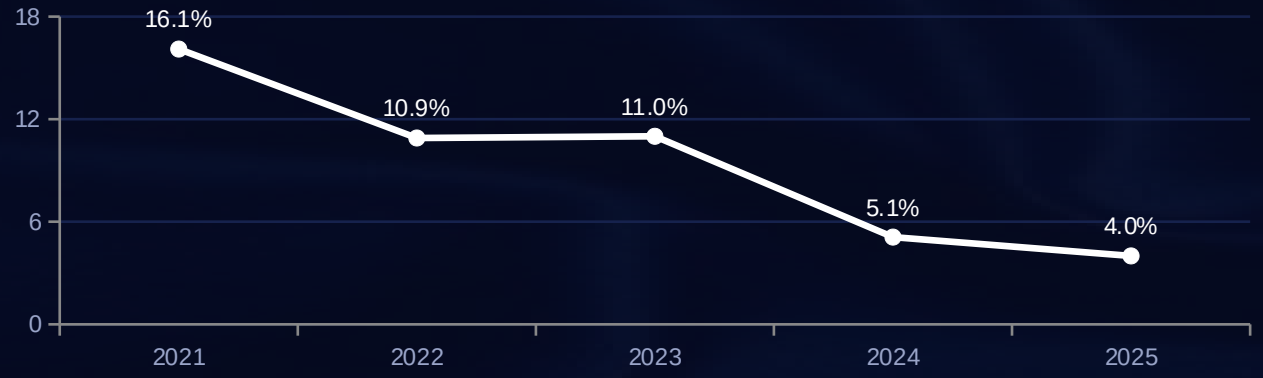
2025: €13.3bn in e-commerce sales — 30.2% of Group sales.

Source: L'Oréal Annual Reports, 2021–2024; 2025 Universal Registration Document.
E-commerce figures are L'Oréal-reported; the underlying KPI uses non-audited self-reported data. E-commerce growth is reported on a like-for-like basis. 2022 and 2023 shares were disclosed as whole percentages.

GROWTH SLOWED WHILE MARGIN KEPT RISING

LFL sales growth fell from 16.1% in 2021 to 4.0% in 2025, while operating margin increased from 19.1% to 20.2%.

LFL SALES GROWTH (%)



OPERATING MARGIN (%)



The latest data therefore shows slowing top-line growth alongside continued margin expansion.

Source: L'Oréal Annual Reports, 2021–2025.

Two metrics moved in opposite directions — this chart does not establish a causal relationship between them.

IF A MEASURED CHANNEL BEHAVES THIS UNEVENLY, WHAT ABOUT AN UNMEASURED ONE?

Across five years, L'Oréal's reported e-commerce data shows that digital growth is not a straight line: share declined for two years before recovering, while overall sales growth slowed even as margin improved.

The next measurement challenge is different. AI can influence discovery and consideration before a customer reaches a retailer or L'Oréal's own channels — creating a new layer of activity that may be harder to observe.

June 2026: L'Oréal announced a partnership with OpenAI covering AI-powered discovery, virtual try-on and advertising initiatives.

Source: L'Oréal, 17 June 2026 VivaTech announcement. Publicly announced scope; L'Oréal-specific measurement implementation is not detailed in public sources.

ATTRIBUTION ≠ INCREMENTALITY

A purchase tagged "AI-referred" is not proof AI caused it.

ATTRIBUTION

AI Discovery → Referral → Retailer Visit → Purchase ✓

This purchase happened, and a tracking signal linked it to an AI referral.

INCREMENTALITY

AI Discovery → Referral → Retailer Visit → Purchase ?

This purchase might have happened even without the AI referral — attribution alone can't tell you.

A tracking signal can attribute a conversion; a comparison-group test is needed to estimate incremental lift.

WHAT WOULD L'ORÉAL NEED TO MEASURE THIS CREDIBLY?

Estimating a credible MDE depends on data and design inputs — here's which ones public data can and can't provide.

Input Needed	Status	Why It Matters
Market-level outcome variance	Not publicly available	Estimates the natural outcome variability an effect must be distinguished from
Eligible comparable test/control markets	Not publicly available	Determines the feasible sample size and comparison structure
Baseline outcome level (e.g. market-level sales)	Not publicly available	Defines the scale against which an incremental lift is evaluated
Sufficient pre-test market history	Not publicly available	Captures baseline trends and seasonality before the intervention
Test duration	Decidable in advance	Determines the observation window and contributes to detectable effect size

No L'Oréal-specific MDE can be calculated from public data. The target effect, alpha and statistical power should be pre-specified as part of the experiment design.



WHAT COULD BE MEASURED — AND WHAT STILL REQUIRES DATA

A proposed framework — not a description of L'Oréal's current measurement systems.

VISIBILITY PROXY	INTERACTION	REFERRAL	CONVERSION	INCREMENTALITY	REPEAT VALUE
AI Answer Mention Rate	**AI-Assisted Session Rate**	**AI Referral Rate**	**AI-Referred Conversion Rate**	**Incremental Sales Lift**	**AI-Originated Repeat Purchase Rate**
Share of standardized test prompts producing a L'Oréal brand/product mention — not real consumer exposure	*Sessions with a verified AI-origin signal reaching a L'Oréal or partner destination*	*Share of AI-originated sessions/referrals identifiable through instrumented links*	*Conversions following an identified AI referral*	*Difference in outcome change between exposed/test markets and matched controls*	*Repeat purchase rate for customers with a reliably identified AI-originated journey*
MEASUREMENT DEPENDENCY					
Standardized prompt testing	Platform and/or referral data	Referral instrumentation	L'Oréal DTC or retailer transaction data	Controlled experiment	Origin-tagged CRM data

WHAT I WOULD TEST FIRST

A feasibility-gated pilot for paid ChatGPT placement.

01

FEASIBILITY CHECK

Confirm whether L'Oréal's specific ChatGPT ad placement — not just the platform's general capability — supports the market-level targeting/assignment and reporting required for the test.

If the required market-level assignment or reporting is unavailable for this partnership specifically: establish the measurement arrangement before running the test.

02

TEST + CONTROL MARKETS

Test markets with paid ChatGPT placement active vs. matched control markets without it.

Market count should be determined from the pre-test power calculation and the eligible market pool.

03

INCREMENTAL SALES LIFT

Compare pre/post sales changes in exposed markets against matched controls. Primary metric: Incremental Sales Lift.

Inconclusive / no effect. If statistical power is insufficient, redesign the test — do not call it zero impact.

04

APPLY DECISION RULE

Predefine the minimum business-relevant lift and decision rule; use the pre-test power calculation to determine whether the design can detect it.

Data required: sufficient pre-test market history · confirmed placement eligibility/assignment · platform/retailer measurement access.

SELF-CORRECTION

I initially framed the pilot around a geo-holdout test. I then checked whether public evidence established the targeting and reporting needed to run one. It didn't — so feasibility became Stage 1 rather than an assumption.

REPORTED SALES GROWTH IS NOT THE SAME AS LFL GROWTH

Two different questions, calculated two different ways, from the same underlying business.

Year	Reported Sales YoY	LFL Sales Growth
2022	18.50%	10.9%
2023	7.64%	11.0%
2024	5.60%	5.1%
2025	1.30%	4.0%

Reported growth includes currency/scope effects; LFL growth is L'Oréal's like-for-like measure. They answer different questions.

Source: L'Oréal Annual Results / reported net sales, 2021–2025.

CAGR AND INDEXING: TWO LENSES, ONE PATH

TOTAL SALES CAGR, 2021–2025

8.08%

E-COMMERCE CAGR, 2021–2025

9.36%

Both CAGRs cover the same 2021–2025 period.

Year	Total Sales Index	E-commerce Index
2021	100.00	100.00
2022	118.50	116.13
2023	127.55	120.43
2024	134.69	132.26
2025	136.44	143.01

Source: L'Oréal Annual Results press releases, 2022–2025 (reported net sales, €m); 2025 Universal Registration Document (e-commerce sales).

A U.S. RETAIL BENCHMARK — CONTEXTUAL ONLY

L'Oréal is global beauty; FRED covers U.S. health & personal-care retail. Different geography, different category.

Year	U.S. Health & Personal Care Retail	L'Oréal LFL
2022	+4.28%	+10.9%
2023	+8.25%	+11.0%
2024	+2.94%	+5.1%
2025	+7.42%	+4.0%

Directional match in 3 of 4 years; contextual only. L'Oréal is global beauty, while FRED covers U.S. health & personal-care retail.

Source: FRED series MRTSSM446USS (U.S. Census Bureau via Federal Reserve Economic Data) — monthly, seasonally adjusted, aggregated to annual.

EUROSTAT: CONSUMER VS. ENTERPRISE ADOPTION

Two sides of digital adoption, 2021 vs. 2025 — contextual background, not a L'Oréal benchmark.

CONSUMER ONLINE-PURCHASE ADOPTION (%)

% of individuals — last online purchase in the last 3 months

Geography	2021	2025	Δ pp
EU-27	56.94	62.35	+5.41
Germany	69.00	70.41	+1.41
France	65.60	70.31	+4.71
Spain	55.22	59.65	+4.43
Italy	39.78	43.79	+4.01

ENTERPRISE B2C WEB-SALES ADOPTION (%)

% of enterprises (10+ employees) with web sales to consumers

Geography	2021	2025	Δ pp
EU-27	15.23	17.17	+1.94
Germany	13.75	16.42	+2.67
France	13.04	14.45	+1.41
Spain	18.17	21.10	+2.93
Italy	13.99	15.29	+1.30

SPAIN, 2024 → 2025: SAME MARKET, OPPOSITE MOVEMENT

Consumer online-purchase adoption: +2.95 pp · Enterprise B2C web-sales adoption: −1.24 pp

Same market, opposite movement — a reminder that consumer demand and business-side digital adoption are not interchangeable measures.

Source: Eurostat isoc_ec_ib20 (consumer online purchases); isoc_ec_esels (enterprise B2C web sales).

WHAT IS ANNOUNCED, DOCUMENTED, AND STILL UNKNOWN

Every AI-related claim used in this project, classified against public evidence.

Platform capabilities are publicly documented; the precise access, eligibility and market-level measurement available to L'Oréal's specific implementation are not publicly established.

Claim	Status
L'Oréal × OpenAI collaboration	ANNOUNCED
AI-powered discovery initiatives	ANNOUNCED
Maybelline Virtual Try-On in ChatGPT	ANNOUNCED
AI advertising initiatives involving L'Oréal brands	ANNOUNCED
ChatGPT Ads geographic targeting	DOCUMENTED
ChatGPT Ads impression/reporting metrics	DOCUMENTED
L'Oréal-specific eligible market assignment for its placement	UNKNOWN
L'Oréal-specific access to the required campaign-level reporting fields	UNKNOWN
Actual consumer exposure volume from organic AI discovery	UNKNOWN
CRM linkage from organic AI discovery	UNKNOWN

Sources: L'Oréal × OpenAI announcement, 17 June 2026; OpenAI ChatGPT Ads documentation, current as of September 2026.

SOURCES & METHODOLOGY

- L'Oréal Annual Reports / Universal Registration Documents, 2021–2025
- L'Oréal 2025 Annual Results
- Eurostat isoc_ec_ib20
- Eurostat isoc_ec_esels
- U.S. Census Bureau via FRED — MRTSSM446USS
- L'Oréal × OpenAI — 17 June 2026 VivaTech announcement
- OpenAI — Create Campaigns for ChatGPT Ads
- OpenAI — Ads in ChatGPT / Measure Results
- OpenAI — Conversion Measurement documentation

Public data only. External datasets are contextual rather than like-for-like benchmarks. Proposed AI measurement constructs are analytical recommendations, not claims about current L'Oréal systems.